



Probability and Statistics

ES-111 Spring 2025

Assignment 3

Deadline of submission: 25.04.2025

Q1.

6.1 Given a continuous uniform distribution, show that

(a) $\mu = \frac{A+B}{2}$ and

(b) $\sigma^2 = \frac{(B-A)^2}{12}$.

Q2.

6.2 Suppose X follows a continuous uniform distribution from 1 to 5. Determine the conditional probability $P(X > 2.5 \mid X \leq 4)$.

Q3.

6.5 Given a standard normal distribution, find the area under the curve that lies

(a) to the left of $z = -1.39$;

(b) to the right of $z = 1.96$;

(c) between $z = -2.16$ and $z = -0.65$;

(d) to the left of $z = 1.43$;

(e) to the right of $z = -0.89$;

(f) between $z = -0.48$ and $z = 1.74$.

Q4.

6.6 Find the value of z if the area under a standard normal curve

- (a) to the right of z is 0.3622;
- (b) to the left of z is 0.1131;
- (c) between 0 and z , with $z > 0$, is 0.4838;
- (d) between $-z$ and z , with $z > 0$, is 0.9500.

Q5.

6.7 Given a standard normal distribution, find the value of k such that

- (a) $P(Z > k) = 0.2946$;
- (b) $P(Z < k) = 0.0427$;
- (c) $P(-0.93 < Z < k) = 0.7235$.

Q6.

6.8 Given a normal distribution with $\mu = 30$ and $\sigma = 6$, find

- (a) the normal curve area to the right of $x = 17$;
- (b) the normal curve area to the left of $x = 22$;
- (c) the normal curve area between $x = 32$ and $x = 41$;
- (d) the value of x that has 80% of the normal curve area to the left;
- (e) the two values of x that contain the middle 75% of the normal curve area.

Q7.

6.11 A soft-drink machine is regulated so that it discharges an average of 200 milliliters per cup. If the amount of drink is normally distributed with a standard deviation equal to 15 milliliters,

- (a) what fraction of the cups will contain more than 224 milliliters?
- (b) what is the probability that a cup contains between 191 and 209 milliliters?
- (c) how many cups will probably overflow if 230-milliliter cups are used for the next 1000 drinks?
- (d) below what value do we get the smallest 25% of the drinks?

Q8.

6.16 In the November 1990 issue of *Chemical Engineering Progress*, a study discussed the percent purity of oxygen from a certain supplier. Assume that the mean was 99.61 with a standard deviation of 0.08. Assume that the distribution of percent purity was approximately normal.

- (a) What percentage of the purity values would you expect to be between 99.5 and 99.7?
- (b) What purity value would you expect to exceed exactly 5% of the population?

Q9:

6.18 The heights of 1000 students are normally distributed with a mean of 174.5 centimeters and a standard deviation of 6.9 centimeters. Assuming that the heights are recorded to the nearest half-centimeter, how many of these students would you expect to have heights

- (a) less than 160.0 centimeters?
- (b) between 171.5 and 182.0 centimeters inclusive?
- (c) equal to 175.0 centimeters?
- (d) greater than or equal to 188.0 centimeters?

- Q10:** **6.23** The IQs of 600 applicants to a certain college are approximately normally distributed with a mean of 115 and a standard deviation of 12. If the college requires an IQ of at least 95, how many of these students will be rejected on this basis of IQ, regardless of their other qualifications? Note that IQs are recorded to the nearest integers.
- Q11:** **5.1.17** Manufactured items have a strength that has a normal distribution with a standard deviation of 4.2. The mean strength can be altered by the operator. At what value should the mean strength be set so that exactly 95% of the items have a strength less than 100?
- Q12:** **5.1.14** The weights of bags filled by a machine are normally distributed with a standard deviation of 0.05 kg and a mean that can be set by the operator. At what level should the mean be set if it is required that only 1% of the bags weigh less than 10 kg?
- Q13:** Assume that in the detection of a digital signal the background noise follows a normal distribution with a mean of 0 volt and standard deviation of 0.45 volt. The system assumes a digital 1 has been transmitted when the voltage exceeds 0.9. What is the probability of detecting a digital 1 when none was sent?
- Q14:** The diameter of a shaft in an optical storage drive is normally distributed with mean 0.2508 inch and standard deviation 0.0005 inch. The specifications on the shaft are 0.2500 ± 0.0015 inch. What proportion of shafts conforms to specifications?
- Q15:** **4-54.** The speed of a file transfer from a server on campus to a personal computer at a student's home on a weekday evening is normally distributed with a mean of 60 kilobits per second and a standard deviation of 4 kilobits per second.
- (a) What is the probability that the file will transfer at a speed of 70 kilobits per second or more?
 - (b) What is the probability that the file will transfer at a speed of less than 58 kilobits per second?
 - (c) If the file is 1 megabyte, what is the average time it will take to transfer the file? (Assume eight bits per byte.)

Q16:

6.45 The length of time for one individual to be served at a cafeteria is a random variable having an exponential distribution with a mean of 4 minutes. What is the probability that a person is served in less than 3 minutes on at least 4 of the next 6 days?

Q17:

6.46 The life, in years, of a certain type of electrical switch has an exponential distribution with an average life $\beta = 2$. If 100 of these switches are installed in different systems, what is the probability that at most 30 fail during the first year?

Q18:

6.62 The exponential distribution is frequently applied to the waiting times between successes in a Poisson process. If the number of calls received per hour by a telephone answering service is a Poisson random variable with parameter $\lambda = 6$, we know that the time, in hours, between successive calls has an exponential distribution with parameter $\beta = 1/6$. What is the probability of waiting more than 15 minutes between any two successive calls?

Q19:

6.66 A certain type of device has an advertised failure rate of 0.01 per hour. The failure rate is constant and the exponential distribution applies.

- (a) What is the mean time to failure?
- (b) What is the probability that 200 hours will pass before a failure is observed?

Q20:

6.81 The length of time between breakdowns of an essential piece of equipment is important in the decision of the use of auxiliary equipment. An engineer thinks that the best model for time between breakdowns of a generator is the exponential distribution with a mean of 15 days.

- (a) If the generator has just broken down, what is the probability that it will break down in the next 21 days?
- (b) What is the probability that the generator will operate for 30 days without a breakdown?